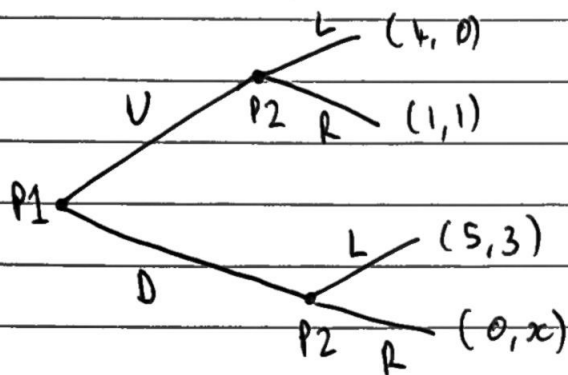


Sheet 4 - dynamic games

i. a)



	LL	LR	RL	RR
U	4, 0	4, 0	1, 1	1, 1
D	5, 3	0, x	5, 3	0, x

where for P2, 'XY' denotes X after U, Y after D.

Good

i.

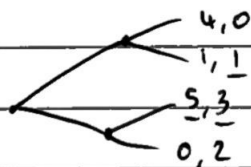
4, 0	<u>4, 0</u>	<u>1, 1</u>	<u>1, 1</u>
<u>5, 3</u>	0, 2	<u>5, 3</u>	0, 2

Underlining BR payoffs, the PSNE are (D, LL), (D, RL) and (U, RR).

iv. know which node they're at

There aren't any MSNE: since P2 has singleton information sets and strict preference at each, they won't mix (otherwise not BR). But then P2 won't mix either, because they know for certain P2's response to U, D and P1 similarly isn't indifferent between any payoffs, so will have a strict preference.

→ There are MNE. Pretend the game isn't sequential and solve for the MNE in the normal form. Using backward induction, we can find SPE is (D, RL).



ii.

4, 0	<u>4, 0</u>	<u>1, 1</u>	<u>1, 1</u>
<u>5, 3</u>	0, 4	<u>5, 3</u>	0, 4

The only PSNE is (U, RR). By same argument as above, no MSNE.

There are MNE.

Since this is a finite horizon game with perfect recall, it has an SPE. And it must also be MNE. So SPE is (U, RR).

iii.

SPE payoffs in i. are (5, 3); in ii. are (1, 1). Perhaps surprisingly, each player is strictly worse-off even though - in a sense - the world only got better: one of the payoffs increased and none

decreased. The problem is of commitment/credibility - it's no longer rational for $p1$ to play D (pulling out the socially-optimal outcome) since in that world they expect $p2$ to rationally play R (to get $4 > 3$) and leave $p1$ with 0 . And $p2$ can't credibly promise to refuse to play R following D since it's the BR.

b) We get a Stackelberg-like game. For $BR_c(x)$, take partials w.r.t strategic variable and set to zero:

$$\partial u_c / \partial y = \frac{(x+y)-y}{(x+y)^2} - 1 \stackrel{!}{=} 0 \Rightarrow BR_c(x) = \sqrt{x} - x$$

Now, substitute this into L 's payoff function for backwards induction:

$$u_L(x, y(x)) = \frac{x}{x + \sqrt{x} - x} - x = \sqrt{x} - x$$

which is maximised by $\partial u_L / \partial x = \frac{1}{2} x^{-1/2} - 1 \stackrel{!}{=} 0 \Rightarrow x = 0.25$

~~and this implies...~~

So the SPE is $(0.25, \sqrt{x} - x)$, which yields exactly the same outcome as the simultaneous game, and indeed the same actions: spending 0.25 each with equal vote shares and 0.25 utility.

But this means there's no first-mover advantage, unlike in sequential Cournot. [Hmm, intuitively why not here?] let's discuss this in class -

2. $D_i = 1 + p_j - p_i$. This is Bertrand competition.

Note $D_i = D_i + D_j = 2$, independent of prices (so in principle there's demand at $p_i = p_j = 0$, which isn't that plausible as a model of consumption)
Hah! Nice point

a) $BR_i(p_j)$ is found by FOC for profits $\pi_i(p_i, p_j) = (1 + p_j - p_i) p_i$
 $\partial \pi_i / \partial p_i = -2p_i + p_j + 1 \stackrel{!}{=} 0 \Rightarrow BR_i(p_j) = \frac{1}{2}(p_j + 1)$
 and similarly $BR_j(p_i) = \frac{1}{2}(p_i + 1)$.

At NE, both play BRs to other's actually-played strategy.
 (p_i^*, p_j^*)

Adding together the two FOCs, we can see

$$p_i^* + p_j^* = \frac{1}{2}(p_i^* + 1) + p_j^* = \frac{1}{2}(p_i^* + 1) + p_i^* \Rightarrow p_i^* = p_j^*.$$

So the unique symmetric NE is $(1, 1)$ with profits $(1, 1)$. No MSNE since only ever strict BRs.

b) $BR_2(p_1) = \frac{1}{2}(p_1 + 1)$. Substituting into payoffs for p_1 , we have

$$\pi_1(p_1, p_2(p_1)) = (1 + \frac{1}{2}(p_1 + 1) - p_1)p_1$$

$$= \frac{3}{2}p_1 - \frac{1}{2}p_1^2$$

which is maximised when $\partial \pi_1 / \partial p_1 = \frac{3}{2} - p_1 \stackrel{!}{=} 0 \Rightarrow p_1 = \frac{3}{2}$. The profit function for p_1 is strictly concave so max is unique.

So the SPE is $(\frac{3}{2}, \frac{1}{2}(p_1 + 1))$; it's unique by concavity.

~~Profits are~~ We end up with prices $(\frac{3}{2}, \frac{5}{4})$, both higher than in simultaneous case. Since demand is fixed, both firms' profits will increase — specifically, we have $(\frac{9}{8}, \frac{25}{16})$.

c) ~~Both~~ Though both firms do better in the sequential setting, the second mover benefits more. In Bertrand, prices are strategic complements, so knowing that the leader has committed to a high price, the follower can undercut slightly while still raising their price above what it'd be in the simultaneous game.

[Is there a systematic relationship between strategic subs. v. compl., and first- v. second-mover advantage? And if so, how come?]
see Bulow & Klemperer

It makes sense that the firms have higher profits when they move sequentially, since this allows for coordination and they'd like to collude to have higher prices.

[though in Cournot, they'd like to collude to have lower quantities, but in fact moving sequentially just increases industry output! Again, I don't have a good intuition for why.]

The first mover has to set high p

d) When simultaneous: $\pi_i = \max\{(1 - q_i - q_j)q_i, 0\}$ with $\Rightarrow$

$$\frac{\partial \pi_i}{\partial q_i} = \begin{cases} 2q_i + 1 - q_j & \text{when } q_i + q_j < 1 \\ 0 & \text{otherwise} \end{cases}$$

Note that it's not a BR to play $q_i = 1 - q_j$ for zero profits, since $1 - q_j - \epsilon$ gives strictly positive profits. So we can just consider the interior ~~solutions~~ points as candidate NE.

$\therefore BR_i(q_j) = \frac{1}{2}(1 - q_j)$. Symmetry is evident from inspecting $q_i^* + q_j^*$.

So at eq* we have $q_i^* = q_j^* = \frac{1}{3}$ with profits $(\frac{1}{9}, \frac{1}{9})$.

In sequential game, substitute in $BR_2(q_1)$ to p1's payoffs:

$$\pi_1(q_1, q_2(q_1)) = (1 - q_1 - \frac{1}{2}(1 - q_1))q_1 = \frac{1}{2}q_1 - \frac{1}{2}q_1^2$$

maximised when $\frac{\partial \pi_1}{\partial q_1} = \frac{1}{2} - q_1 \stackrel{!}{=} 0 \Rightarrow q_1 = \frac{1}{2}$

So SPE is $(\frac{1}{2}, \frac{1}{2}(1 - q_1))$ with ~~product~~ industry output now $\frac{1}{2} + \frac{1}{4} = \frac{3}{4} > \frac{2}{3}$ ^{vs} the simultaneous setting. Industry profits are

hence lower ^{in total} ~~overall~~, but the first-mover has an advantage, because $\hookrightarrow (\frac{1}{2}, \frac{1}{4})$ they can increase production ~~which~~ ^{which} due to quantities being

~~direct effect of more market share and indirect via~~ strategic substitutes, ~~of~~ reducing ^{at} p2's output.

3. a)

	A	B
A	(<u>3</u> , 1)	(0, 0)
B	(0, 0)	(1, <u>3</u>)

PSNE by underlining BR payoffs, or just inspection: (A, A) and (B, B).

^{There's unique} You can't have only one player mixing because a BR to pure strategy from opponent due to strict ⁱⁿeq. between payoffs.

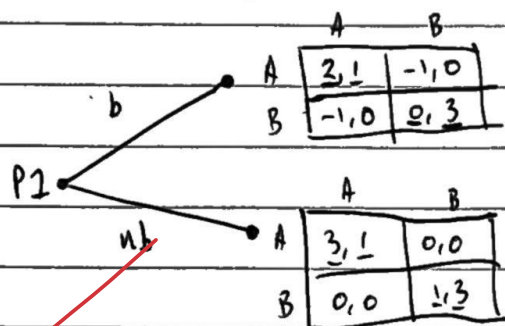
Let p, c play A w.p. p ~~q~~ ^p, q. Then by indifference principle we have

$$v_r(A) = v_r(B): 3 \cdot q = 1 \cdot (1-q) \Rightarrow q = 1/4$$

$$v_c(A) = v_c(B): 1 \cdot p = 3 \cdot (1-p) \Rightarrow p = 3/4$$

So MSNE is $(\frac{3}{4}A + \frac{1}{4}B, \frac{1}{4}A + \frac{3}{4}B)$
with $\mathbb{E}$ payoffs $(\frac{3}{4}, \frac{3}{4})$

b)



	AA	AB	BA	BB
bA	2, 1	2, 1	-1, 0	-1, 0
bB	-1, 0	-1, 0	0, 3	0, 3
nbA	3, 1	0, 0	3, 1	0, 0
nbB	0, 0	1, 3	0, 0	1, 3

Where for $p2$, 'XY' denotes X if $p1$ burns, Y if not.

c) Can find PSNE by underlining BR payoffs in strategic form.
We have (nbA, AA) , (nbA, BA) , (bA, AB) , (nbB, BB)

Can find SPE by backwards induction. In b subgame, PSNE are (A, A) and (B, B) ; same in nb subgame but payoffs $(3, 1)$ and $(1, 3)$.

Yes!!
 This is in fact why I don't like the fact that the question tells you to look at reduced strategies

[One thing that seems confusing here: if we look at e.g. (nbA, AB) , it doesn't tell us what $p2$ would've done had they been in the burn subgame, so it's unclear how to determine if the profile is NE in that subgame. There is some NE in that subgame where $p2$ plays B , though, and maybe that's all we require?]

↳ who does have an action specified

(nbA, AA) : ~~is a subgame~~ NE in nb subgame, and b subgame, and $p2$ chooses subgame with higher payoff $(3, 1)$ vs $(2, 1)$. ✓
 (nbA, BA) similar though $(3, 1)$ vs $(0, 3)$ ✓
 (bA, AB) similar, $(2, 1)$ vs $(1, 3)$ ✓
 (nbB, BB) similar, $(1, 3)$ vs $(0, 3)$ ✓ So all are SPE ✓

	AA	AB	BA	BB
bA	2, 1	2, 1	-1, 0	-1, 0
bB	1, 0	1, 0	0, 3	0, 3
nbA	3, 1	0, 0	3, 1	0, 0
nbB	0, 0	1, 3	0, 0	1, 3

Let ' $\leq$ ' denote weak dominance by.

(1) $bB \leq nbA$ (2) $BB \leq AB$ (3) $nbB \leq nbA$

(4) $AB \leq AA$ and $BA \leq AA$ (5) $nbA \leq nbA$

So the IEPDS-surviving strategies ^{profile} is (nbA, AA), which is SPE as established.

e) ~~reason~~ apply the Iterative Elimination

If p_1 chooses to burn their money, then they lose 1 utility in each outcome, compared to not burning. So it could only be rational for them to do this if they thought it'd change the outcome arrived at.

[The idea is that p_1 ^{would} signal to p_2 that they're set on playing A by burning? But I don't see how this means we end up in the non-burn subgame, or why p_2 won't play B given we're there (+ not in the burn one).]

It seems somewhat unlikely to me that this is a good prediction of what would happen in real life. Reasoning through forward or backward induction is cognitively difficult, so agents might not be able to compute that ~~one~~ only one SPE survives in this way. Also, since we in reality don't have common knowledge of rationality, observing the opponent burning money increases p_2 's belief that p_1 is crazy, rather than interpreting this purely as a signal from a rational agent, so the forward induction doesn't necessarily go through.

→ Fundamentally, how is it different from backward induction?